

# Physiological and immunological responses of birds and mammals to forest degradation: A meta-analysis

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## ABSTRACT

Dramatic changes in species composition have been found following selective logging and forest fragmentation. The different responses of bird species to these disturbances suggest that some species are more sensitive to environmental changes than others. Recent studies have suggested that the chances of species to adapt to new environments may be mediated by their stress physiology and immunity. We reviewed and performed a meta-analysis of studies that compared physiological and immunological endpoints of bird and mammal species between degraded (logged and fragmented) forests and undisturbed forests. We found that stress hormones and immunity markers show consistent changes in response to habitat degradation in birds and mammals. Higher physiological and immunological responses were found in those animals living in forests that were subjected to clear-cutting. Furthermore, we found that birds and species belonging to IUCN 'Threatened' categories exhibited significantly larger effect size estimates than mammals and 'Least Concern' species, respectively. Our meta-analysis revealed that changes in the production of stress hormones and in some immune traits are a significant consequence of forest disturbance. Physiology and immunity might be two important mediators of the adaptiveness of a given species to changing forests.

## 1. Introduction

Land-use change is a key driver of the global extinction crisis, driving losses in biodiversity and functional diversity (Newbold et al., 2015; Tilman et al., 2017). Selective logging and conversion to agricultural lands or pastures, fragmenting natural habitat and increasing edge effects, are the main processes by which humans are modifying natural landscapes worldwide (Fletcher et al., 2007; Lewis et al., 2015). Such habitat changes alter abiotic conditions and biotic interactions, leading to changes in species' population and community structure (Bogdziewicz and Zwolak, 2013; Fahrig, 2002; Perry and Thill, 2013; Ries and Sisk, 2004).

Although primary forests and strictly protected areas are vital to achieve global biodiversity goals (Chaudhary et al., 2016; Laurance et al., 2012), their relatively small areal extent means that logged and secondary forests are increasingly recognised to be essential to preserve global biodiversity (Edwards et al., 2014). In particular, mature logged forests retain a similar species richness to that of primary forests and far higher than that of agricultural and agroforestry systems (Gibson et al., 2011; Newbold et al., 2015; Putz et al., 2012). Protecting logged forests

from land-use change thus provides a valuable contribution to slowing biodiversity loss (Edwards et al., 2011; Lewis et al., 2015).

Species adaptiveness to environmental changes may be determined by a number of factors, including ecological, functional or morphological traits. For example, Hamer et al. (2015) found that bird species from a single-source feeding guild (i.e., dietary specialists) are more prone to decline in selectively logged forests than species that use two or more groups of food. Other studies observed that different forest dependent bird and mammal species, such as cavity-nesting (e.g., pine marten: *Martes martes*; white-backed woodpecker *Dendrocopos leucotos*) and larger-bodied (e.g., northern spotted owl *Stryx occidentalis*; baird's tapir *Tapirus bairdii*) species, are particularly sensitive to clear-cutting and fragmentation (Czeszczewik and Walankiewicz, 2006; Fisher and Wilkinson, 2005; Jordan et al., 2016; Wasser et al., 1997). By contrast, a recent study investigating edge sensitivity of endotherms found lower edge-sensitivity in large compared to small-to-medium-sized forest mammals, and minimal evidence supporting a positive relationship between edge-sensitivity and body size in birds (Pfeifer et al., 2017).

Although we now have a better understanding of the functional and morphological traits that determine whether a species is going to

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decline or succeed in degraded environments (Edwards et al., 2013; Gray et al., 2007; Pryde et al., 2016), there remains a need to understand the physiological mechanisms that determine how organisms respond and adapt to environmental alterations (Wingfield, 2013). A recent meta-analysis of the impact of selective logging on tropical birds and mammals concluded that there are unidentified explanatory variables that determine a species' response to logging and pointed to the importance of physiology as a mediator of the consequences of land-use change (Costantini et al., 2016).

The study of physiological responses of vertebrates to habitat changes has an extraordinary potential to recognize environmental stressors and to quantify their potential impact on individual reproductive fitness (Coristine et al., 2014; Wikelski and Cooke, 2006). The stress response of vertebrates to environmental stimuli is mediated by glucocorticoid hormones, which are the end products of a hormonal cascade modulated by the hypothalamic-pituitary-adrenal (HPA) axis in response to stress exposure (Romero, 2004). Quantification of glucocorticoids in biological matrices (e.g., plasma, feces, feathers) is one of the most widely used approaches to assess the physiological stress state of organisms (Johnstone et al., 2012b). For instance, Suorsa et al. (2003) showed that Eurasian treecreeper (*Certhia familiaris*) chicks from young regrowth forests exhibit higher plasma corticosterone (CORT) than chicks from sparse, old forests.

Habitat alterations can also affect the immune system, hematological parameters and parasite load. Immunosuppression resulting from habitat changes may represent an adaptive form of energy savings or a direct effect of allostatic overload (Martin, 2009; McEwen and Wingfield, 2003). Irwin et al. (2010) found that diademed sifakas (*Propithecus diadema*) inhabiting fragments of selectively logged forest exhibit lower levels of white blood cells (WBC) and packed cell volume (PCV) compared to individuals living in undisturbed forests. Renner et al. (2016) found that blackcaps (*Sylvia atricapilla*) and chaffinches (*Fringilla coelebs*) were more likely to acquire an infection in less structured forests, even if they exhibited different immune responses.

In this article we have reviewed, for the first time, studies that compared physiological and immunological endpoints of bird and mammal species between degraded (logged and fragmented) forests and undisturbed forests. We conducted a meta-analysis to estimate the size of the effects of logging and fragmentation and we assessed whether the effect size differs between birds and mammals and between species with different International Union for Conservation Nature (IUCN) Red List status.

## 2. Materials and methods

### 2.1. Data collection

We focused on studies that compared glucocorticoids, immune endpoints and hematology (Table A1, Appendix B) of mammals and birds between disturbed and undisturbed (control) forests. As types of forest disturbance we considered clear-cutting, selective logging, fragmentation and edge effects. We considered only studies where selective logging occurred within 10 years of data collection. Forest fragments frequently visited for ecotourism were not taken into account because tourism can affect species' stress responses (Müllner et al., 2004; Shutt et al., 2014). Reptiles and amphibians have been excluded due to the scarcity of results found in the literature (1 and 3 studies respectively matched our search criteria). We performed a review of the literature on the Web of Knowledge using combinations of the keywords “forest log\*”, “clear cutting”, “forest disturb\*”, “forest fragm\*”, “forest edge”, with “glucocort\*”, “corticostero\*”, “cortisol”, “parasit\*”, “stress physiol\*” and “health parameter\*”. The last search was conducted on 18 September 2017. This search resulted in the screening of approximately 2100 articles.

From each article that matched our search criteria, we collected sample sizes and statistical outcomes (mean, standard error or standard

deviation or 95% confidence interval) of the response parameters considered, in undisturbed and disturbed forests. If data were not reported in the text or tables, we extracted them from graphs using the software GetData Graph Digitizer (Fedorov, 2014). When extraction of data was not possible, the authors were contacted. Twelve out of 16 authors responded (Appendix B) and eight of them provided data which were used to calculate the effect size.

### 2.2. Effect size calculation

We calculated effect size Hedges' *g* from test statistics collected from selected papers using the *compute.es* package (Del Re, 2013) in R (R Core Team, 2013). Hedges' *g* is the result of the multiplication of Hedge's *d* (i.e., standardized mean difference) with a correction factor calculated from the degrees of freedom. This conversion is necessary due to the small within-study sample sizes (Cooper et al., 2009; Smith et al., 2016). Our work included 170 effect size estimates from 28 studies (Table A1, Appendix B) (Aronsen et al., 2015; Arriero et al., 2008; Balestri et al., 2014; Butler et al., 2013; Cantarelli et al., 2017; Carlitz et al., 2016; Chapman et al., 2006; Cottontail et al., 2009; Gillespie and Chapman, 2008; Grava et al., 2013; Irwin et al., 2010; Jaimez et al., 2012; Johnstone et al., 2011, 2012a; Leshyk et al., 2012; Leshyk et al., 2013; Lucas et al., 2006; Martínez-Mota et al., 2007; Mastromonaco et al., 2014; Mazerolle and Hobson, 2002; Mborra et al., 2009; Rakotoniaina et al., 2016; Rangel-Negrin et al., 2014; Rivers et al., 2013; Santicchia et al., 2015; Trejo-Macias et al., 2007; Wasser et al., 1997; Wells et al., 2007). For the statistical analyses, all estimates of effect size were made positive because we were interested in the magnitude of the effect rather than in the direction. Effect sizes were considered to be small (Hedges *g* = 0.2, explaining 1% of the variance), intermediate (*g* = 0.5, explaining 9% of the variance) or large (*g* = 0.8, explaining 25% of the variance) as suggested by Cohen (1988).

### 2.3. Meta-analysis

The effect of logging on physiological and immunological parameters can be influenced by various factors. For this reason we included in the analysis some explanatory variables (termed moderators in meta-analysis). An important moderator is the ‘Disturbance’ occurring in the forests and this moderator included ‘Clear Cutting’, ‘Selective Logging’, ‘Fragmentation’ and ‘Edge Effect’. Selective logging and clear-cutting occurred in tropical and temperate biome, respectively (only two studies reported selective logging in temperate forests, none reported clear cutting in the tropics), and thus to avoid redundancy we did not include biome as moderator in the meta-analysis (MA).

To assess whether the effect of disturbance varies among physiological endpoints, we also included in the analyses the moderator ‘Parameter’ categorized as follows: (1) ‘Hormonal Response’ (e.g., corticosterone and cortisol measured in scats, integument or blood); (2) ‘Immunity’, which included white blood cell counts (WBC) and relative presence of different WBC types; (3) ‘Parasite-load’ measured in blood, feces or ectoparasites; and (4) ‘Hematology’ including a range of reference blood values (e.g., electrolytes, enzymes, vitamins).

Stress hormones measured in different substrates or following a capture-restraint protocol may change in their intensity of response to disturbance. Thus, we tested whether there were differences in effect size among the subcategories of ‘Hormonal Response’: ‘Baseline’ (corticosterone or cortisol measured in plasma collected within 3 min from the capture), ‘Stress Induced’ (corticosterone or cortisol measured in plasma collected after a restraint protocol), ‘Short-term’ and ‘Long-term’ (corticosterone and cortisol measured in scats and integument (fur or feathers), respectively).

We finally tested for the moderator ‘Class’ (‘Birds’ vs. ‘Mammals’ or ‘Birds’ vs. ‘Primates’ vs. ‘Other Mammals’, because primates were the most representative taxa within mammals) and the moderator ‘IUCN’, which indicates the conservation status of the study species. We

considered ‘Least Concern’ category versus ‘Threatened’ category (near threatened, vulnerable, endangered and critically endangered pooled together).

## 2.4. Meta-analytic technique

The *rma.mv* function of the *metafor* package of R (R Core Team, 2013) was used to implement our meta-analytic multilevel mixed-effects models. To test the absence of publication bias we performed funnel plot analysis of ‘unsigned *g*’, and to quantify the bias we calculated the fail-safe *N* using Rosenthal approach (Rosenthal, 1979). The fail-safe *N* represents the minimum number of additional studies that would be needed to overturn the results of the meta-analysis. We used the Hedges’ *g* values weighted for their variances (calculated using *compute.es*) as response variables in our statistical models. Estimates of effect size derived from the same article or from the same species are not independent, thus to control for pseudoreplication we included as random factors the article ‘ID’ and the ‘Species’. The factors ‘Class’ and ‘Parameter’ were also included as random factors in the models; they were retained in the models whether they improved the fitting of the model tested by Akaike Information Criterion with correction for small sample sizes. The *QE*-test for residual heterogeneity was performed to test whether there is significant unexplained variance not accounted for by the moderators included in the model. The *QM*-test of model coefficients was performed to test the effects of moderators and predictions of mean and 95% confidence interval of single moderators were calculated.

## 3. Results

The symmetrical shape of our funnel plot indicates the absence of publication bias in our dataset (Fig. A1, Appendix A). Moreover, the fail-safe number was very large ( $N = 23,725$ ).

Unsigned effect size estimates were generally small to intermediate according to Cohen (1988). The effect of forest disturbance on species’ physiology and immunology was significantly larger for ‘Birds’ compared to ‘Mammals’ ( $Q_M = 5.64$ ,  $df = 1$ ,  $P = 0.0176$ ) and for ‘Threatened’ species compared to ‘Least Concern’ species ( $Q_M = 10.39$ ,  $df = 1$ ,  $P = 0.0013$ ). To reduce differences in sample size between ‘Birds’ and ‘Mammals’ categories (see Table A1, Appendix B), we split the latter in ‘Primates’ and ‘Other Mammals’, respectively. The effect of category was still significant ( $Q_M = 8.3287$ ,  $df = 2$ ,  $P = 0.0155$ ; Figs. 1 and 3) with both mammal categories exhibiting lower effect size estimates than birds (‘Primates’: mean = 0.36, CI = 0.26–0.47; ‘Other Mammals’: mean = 0.25, CI = 0.12–0.38; ‘Birds’: mean = 0.52, CI = 0.37–0.66).

The effect size of forest disturbance significantly differed among parameters ( $Q_M = 9.41$ ,  $df = 3$ ,  $P = 0.0243$ ). The size of effect (i.e.,

response to disturbance) was larger for immunity (mean = 0.42, CI = 0.29–0.56) and stress hormones (mean = 0.44, CI = 0.32–0.55) than for hematology (mean = 0.31, CI = 0.18–0.45) and parasites load (mean = 0.31, CI = 0.21–0.42; Fig. 2). A pairwise comparison, however, highlighted significant differences between immunity and hematology only (Table 1). The subcategories of ‘Hormonal responses’ did not significantly differ in effect size ( $Q_M = 3.25$ ,  $df = 3$ ,  $P = 0.3546$ ).

Non-significant differences were found for overall forest disturbances ( $Q_M = 6.54$ ,  $df = 3$ ,  $P = 0.0881$ ). However, clear-cutting (mean = 0.52, CI = 0.36–0.69) showed a larger effect size compared to other disturbances (Fig. 2). Significant differences emerged between clear-cutting and fragmentation (mean = 0.33, CI = 0.22–0.44) or edge-effect (mean = 0.25, CI = 0.07–0.43; Table 1), respectively.

The significant levels of residual heterogeneity revealed by *QE*-test (Table 2) indicated that moderators used in this meta-analysis were not sufficient to explain all the variance.

## 4. Discussion

Fragmented and selectively logged forests now dominate globally and can harbour high levels of biodiversity compared to agricultural fields, but how individual species respond physiologically to disturbance remains an open question. Stress physiology (e.g., glucocorticoid hormones) mediates a rapid organismal response to environmental changes before genetic adaptations occur and such responses may have an important role in determining the chances of a species to adapt to new environments. Using a meta-analytical approach, we found that stress hormones and immunity markers show consistent changes in response to habitat degradation in birds and mammals (Fig. 1). Furthermore, stronger physiological and immunological responses were found (i) in animals living in forests that were subjected to clear-cutting (Fig. 2), which is the most destructive method of logging, and (ii) in those species that are currently classified as threatened by IUCN. Finally, we also found that in birds the effect size of physiological and immunological responses to habitat degradation was higher than that of mammals, even if the among species variation in effect size estimates was high (Fig. 3).

Mean effect size estimates in our MA were generally small to intermediate, in line with findings in ecological and physiological research (e.g., Dantzer et al., 2014; Möller and Jennions, 2002). This result might be due to the fact that all studies included in our MA were conducted in the wild, where many sources of variation could not be taken into account. However, small to intermediate effect size are thought to be biologically meaningful when showing consistent patterns of variation (Garamszegi, 2006), as shown by most of the 95% confidence intervals of the predicted variables included in this MA.

The use of unsigned effect size in our MA allowed us to find a general marked response of stress hormones in those animals living in

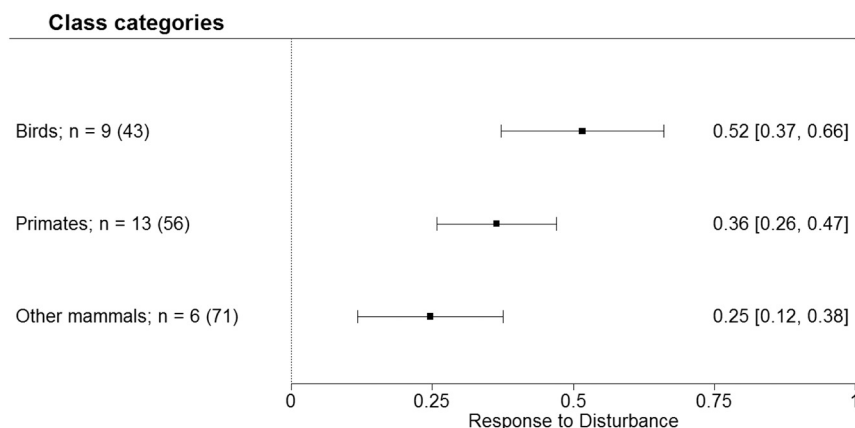


Fig. 1. Predicted unsigned effect sizes (mean and 95% confidence interval) for Class categories. ‘n’ is number of studies (number of effect sizes).

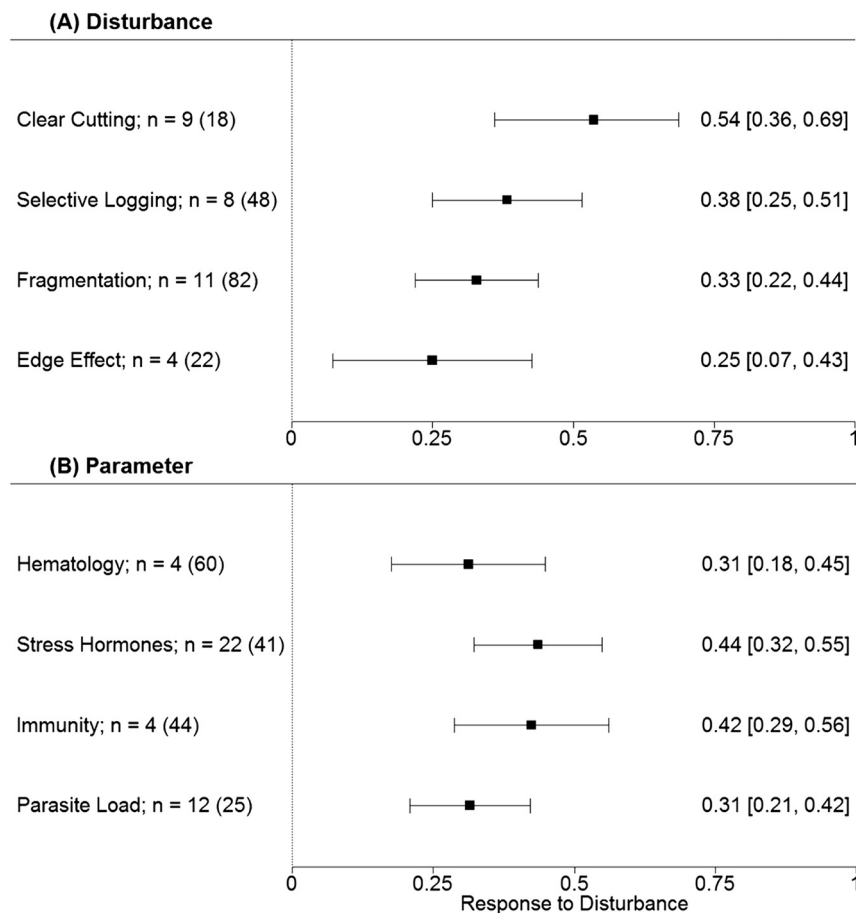


Fig. 2. Predicted unsigned effect sizes (mean and 95% confidence interval) for Disturbance and Parameter categories. 'n' is number of studies (number of effect sizes).

Table 1

Pairwise comparison between effect sizes for (A) 'Disturbance' and (B) 'Parameter' categories.

| Pairwise comparison               | Estimate | Standard error | P value |
|-----------------------------------|----------|----------------|---------|
| <b>A. Disturbance</b>             |          |                |         |
| Clear cutting – selective logging | –0.142   | 0.098          | 0.1504  |
| Clear cutting – fragmentation     | –0.200   | 0.092          | 0.0328* |
| Clear cutting – edge effect       | –0.274   | 0.117          | 0.0193* |
| Selective logging – fragmentation | –0.054   | 0.075          | 0.4742  |
| Selective logging – edge effect   | –0.132   | 0.106          | 0.2146  |
| Fragmentation – edge effect       | –0.08    | 0.099          | 0.431   |
| <b>B. Parameter</b>               |          |                |         |
| Hematology – hormones             | 0.14     | 0.09           | 0.1213  |
| Hematology – immunity             | 0.112    | 0.044          | 0.011   |
| Hematology – Parasites            | –0.002   | 0.07           | 0.98    |
| Hormones – immunity               | 0.028    | 0.09           | 0.7595  |
| Hormones – parasites              | –0.138   | 0.08           | 0.0869  |
| Immunity – parasites              | –0.1103  | 0.07           | 0.1157  |

Table 2

QE-test for residual heterogeneity.

| Model                                      | QE      | d.f. | P value |
|--------------------------------------------|---------|------|---------|
| Disturbance                                | 212.949 | 166  | 0.0081* |
| Class (birds vs mammals)                   | 220.818 | 168  | 0.0039* |
| Class (birds vs primates vs other mammals) | 213.609 | 167  | 0.0086* |
| Parameter                                  | 232.804 | 166  | 0.0005* |
| IUCN                                       | 226.148 | 168  | 0.0019* |

disturbed forests. The high responsiveness of stress hormones indicates the implication of the HPA axis in the mechanisms of adaptation of bird and mammal populations to habitat changes (consequent to the regulation of glucocorticoids level). We chose to not investigate the direction of the physiological response (i.e., increase or decrease) for three main reasons: 1) habitat degradation may also induce chronic stress that could lead to HPA axis hypo-activity with a consequent reduction in baseline concentration of stress hormones (Dickens and Romero, 2013); 2) different species perceive new environments differently and some species may benefit from new conditions created by habitat changes. For example, Aronsen et al. (2015) found significantly higher cortisol concentrations in two species of monkeys (*Cercopithecus ascanius* and *Ptilocolobus rufomitratus*) living in undisturbed forests compared to disturbed sites; and 3) the lack of data on fitness traits complicates interpretation of the direction of the change, e.g. if this is indicating a detrimental effect on reproduction.

Glucocorticoids can induce a number of cascade effects on metabolism, such as the mobilization of stored nutrients to be used to fuel other physiological functions, including immunity or resistance to molecular oxidative damage (Costantini et al., 2011; Marasco et al., 2017). Although our MA supports a significant impact of forest disturbance on immunity, we did not find any studies that looked at the impact of land-use changes on oxidative damage resistance. As previously suggested by other authors (Beaulieu and Costantini, 2014; Davis et al., 2008; Müller et al., 2011), we recommend that future studies would complement analyses of stress hormones with those of immunity and oxidative stress status in order to have a clearer picture of how land-use changes are affecting a given species and how this species is responding.

Our results also show that clear cutting has a stronger impact than

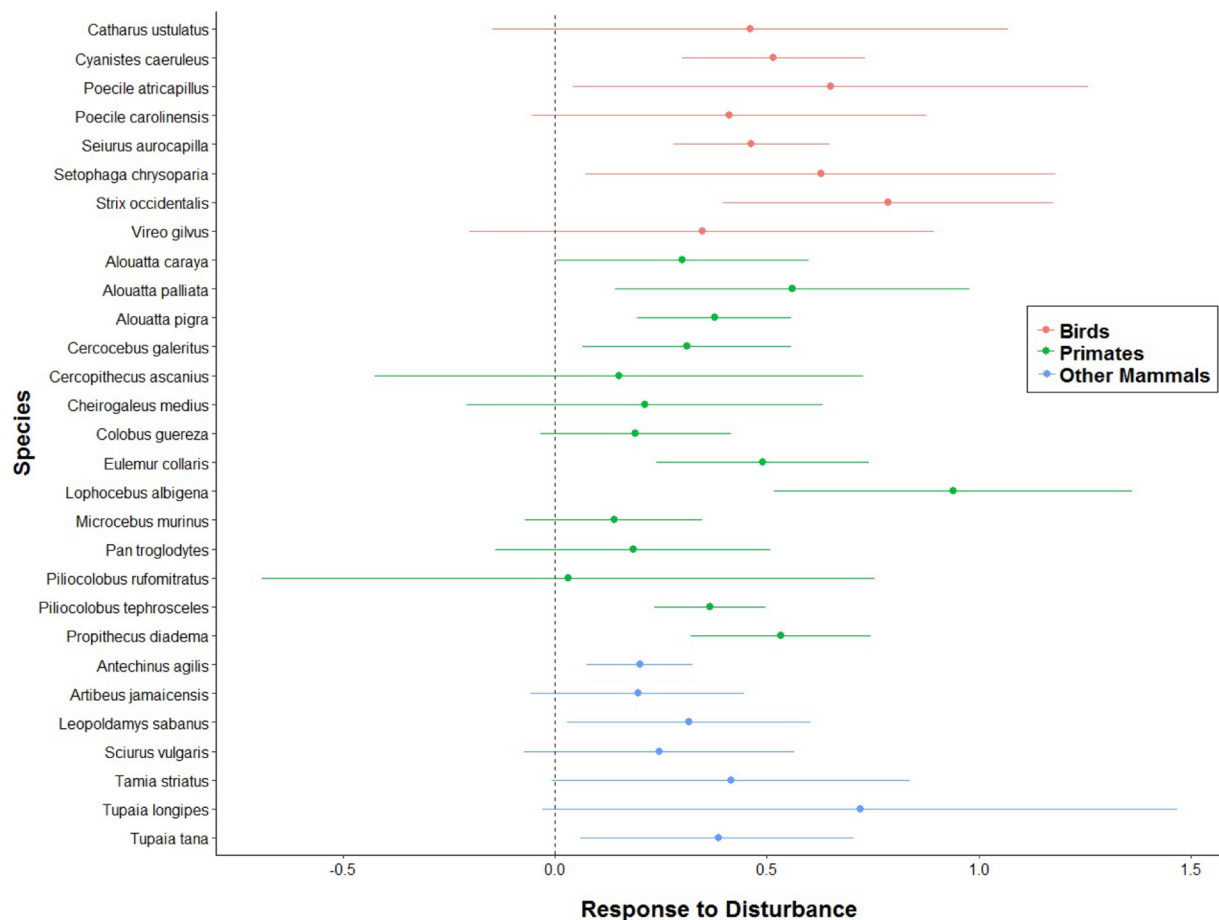


Fig. 3. Bird, primate and other mammal species response (unsigned mean ES and 95% confidence interval) to forest disturbance.

selective logging, fragmentation or edge effect. Clear-cutting and selective logging differ markedly in their impact on the vegetation and the amount of timber that is being removed from the forest (Woodcock et al., 2015). Several studies found a limited effect of selective logging on vertebrate species richness (Chaudhary et al., 2016; Edwards et al., 2011; Gibson et al., 2011; Putz et al., 2012), while major changes in bird and mammal communities were observed in post clear-cutting environments (Fisher and Wilkinson, 2005; Fraixedas et al., 2015). We could not include important variables in our MA, such as the intensity of logging (Burivalova et al., 2014), the size of forest fragments (Arroyo-Rodríguez et al., 2013) and the distance to forest edge (Ewers and Didham, 2006), which might have a strong impact on the species' physiology and immunology. Thus we highlight the importance of testing the effects of logging intensity, fragment size and distance to forest edge on the physiological and immunological status of vertebrates in future studies.

We found a larger effect of forest disturbance on birds than on primates or other mammals. This result might be affected by the high number of studies on birds inhabiting clear-cut forests included in the MA. It is well known that forest stand homogeneity in structure and composition limits the availability of nest sites and canopy cover (e.g., Müller et al., 2007; Rosenvald et al., 2011) and reduces sources of food for many bird species (Paillet et al., 2010), with potential negative effects on individual fitness and survival. In contrast, results of studies investigating physiological responses to habitat changes in group-living primates may be influenced by inter- and intra-group social dynamics (Aronsen et al., 2015; Chapman et al., 2006), which can have a strong influence on individual health (Sapolsky, 2005).

Recent studies found that logging or fragmentation may differentially affect species with different conservation statuses (IUCN category;

Boyle, 2016; Costantini et al., 2016). For example, a study conducted on the two sympatric species, the 'least concern' red howler monkey (*Alouatta seniculus*) and the 'critically endangered' variegated spider monkey (*Ateles hybridus*) found that only the latter had higher levels of stress hormones due to anthropogenic disturbance (Rimbach et al., 2013). In agreement with this study, we also found that species Red listed by the IUCN as threatened or near threatened are more affected by forest disturbance than species classified as of least conservation concern.

Variation in effect sizes in our MA might be affected by a number of factors that we could not take into account because of lack of data. Some limiting factors of our results are pointed out and discussed in the paragraphs above, but sources of variation might be many and different. For example, sex and age were excluded from our MA because in most studies they were mixed or unknown. Similarly, different diet specialization and body size may affect the species sensitivity to forest disturbance (Keinath et al., 2017). Moreover, given the broad array of physiological and immunological parameters and samples analysed, it has not been possible to estimate the effect size of each parameter separately. We could only test the effect size of subcategories of Hormonal Response (Baseline vs. Stress Induced vs. Long-term vs. Short-term), and we did not find any significant differences between them.

In conclusion, our MA revealed that changes in the production of stress hormones and in some immune traits are significant consequences of forest disturbance, particularly of clear-cutting. The significant association between threatened IUCN category and higher physiological response suggests that physiology and immunity might be two important mediators of the (mal)adaptiveness of a given species to the changing forests. Our work has also identified many gaps in current literature and stresses the importance of experimental approaches

aimed to better clarifying the role of physiology and immunity in determining the chances of species to survive or disappear in human-impacted environments.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biocon.2018.06.002>.

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